EXPERIMENT 12

Implementing Security FeaturesandGovernance / Administration Practices in Zoho Creator

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Subject: Low Code Application Development using Zoho Creator

Aim

To implement and explore the security, access-control, data-sharing and governance features of the Student Admission application in Zoho Creator.

Introduction

Security and governance in low-code applications deal with controlling who can use an application, what each person is permitted to do, and how the data and resources of the application are administered. Because low-code platforms make it easy to build applications that store business data, controlling access to that data is an important part of application design. This is especially true for an admissions application, which holds personal details, marks and documents of applicants.

Zoho Creator provides built-in mechanisms for this purpose. User permissions define the operations that a user may perform on the components of an application. Role-based access places users in roles that are arranged in a hierarchy, which influences the records that they can see. Data sharing rules extend the visibility of records from one role or user to another with a defined level of access. Administrative controls allow the application to be configured and

managed. Together, they help ensure that users receive only the access needed for their responsibilities and that application data is protected from unauthorised use.

Application Used

The practical was carried out in Zoho Creator using the Student Admission application. The application is used for managing student applications and admissions, and it served as the practical environment for exploring the security and governance features described in this record.

Security and Governance Architecture

Figure 1 is a conceptual supporting diagram of the security and governance structure of the Student Admission application. It is not a screenshot of the Zoho Creator interface.

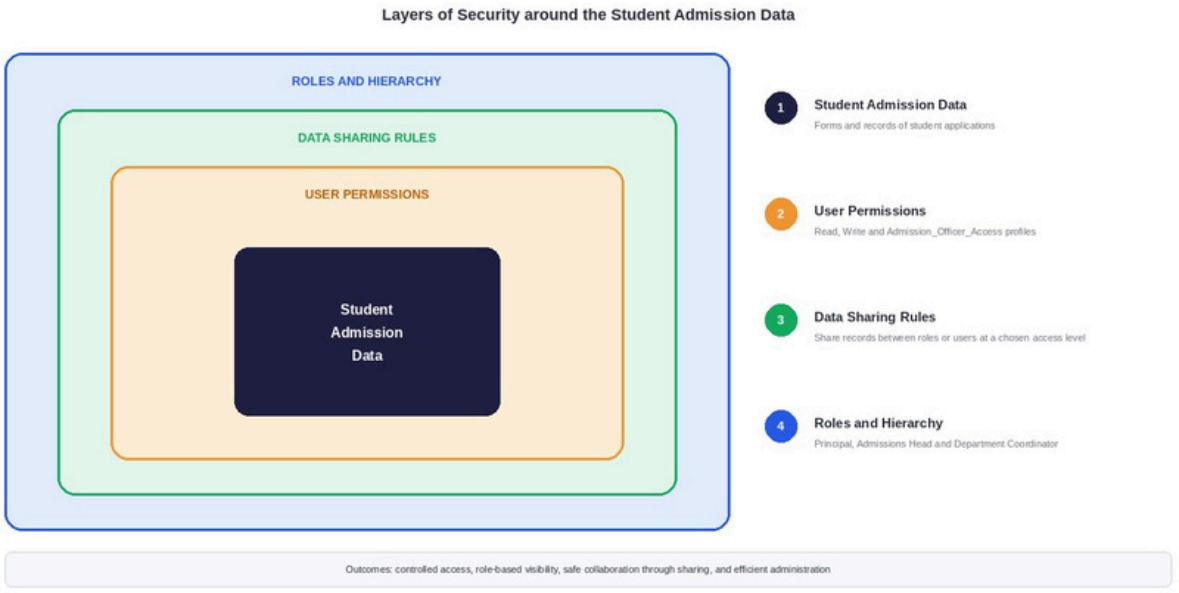


Figure 1: Security & Governance Layers (Conceptual Diagram)

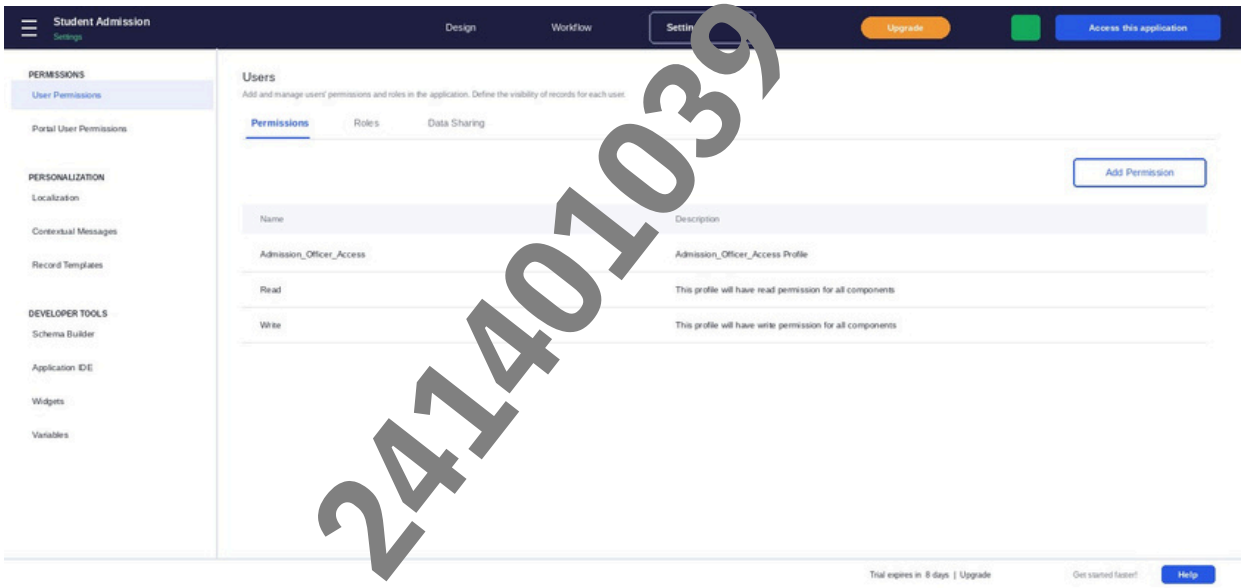
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The diagram shows the admission data of the application at the core, protected by successive layers. User permissions (Read, Write and Admission_Officer_Access) define what a user may do. Data sharing rules extend the visibility of records between roles or users at a chosen level of access. The role hierarchy of Principal, Admissions Head and Department Coordinator forms the outermost layer and influences which records each role can see. The key outcomes are controlled access to data, role-based visibility, safe collaboration through data sharing, and efficient administration.

User Permissions

The User Permissions section is opened from Settings in the application editor. It contains the tabs Permissions, Roles and Data Sharing, and is used to add and manage the permissions and roles of users and to define the visibility of records for each user. The Permissions tab lists the permission sets available in the application and provides an Add Permission option. The application listed three permissions:

Name	Description (as displayed)
Admission_Officer_Access	Admission_Officer_Access Profile
Read	This profile will have read permission for all components
Write	This profile will have write permission for all components



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Figure 2: User Permissions in the Student Admission application

Read permission allows a user to view the components and records of the application without changing them, whereas Write permission allows the user to make changes, such as adding or modifying records. Admission_Officer_Access is a permission profile named for the admission officers of the institution; it is listed alongside the Read and Write profiles and can be assigned to users who process student applications.

Role-Based Access Control

The Roles tab shows the role hierarchy of the application. In the Student Admission application the hierarchy is displayed with Principal at the top, Admissions Head below it and Department Coordinator below that. The options Expand All, Collapse All, Manage Role Hierarchy and Add Role are available.

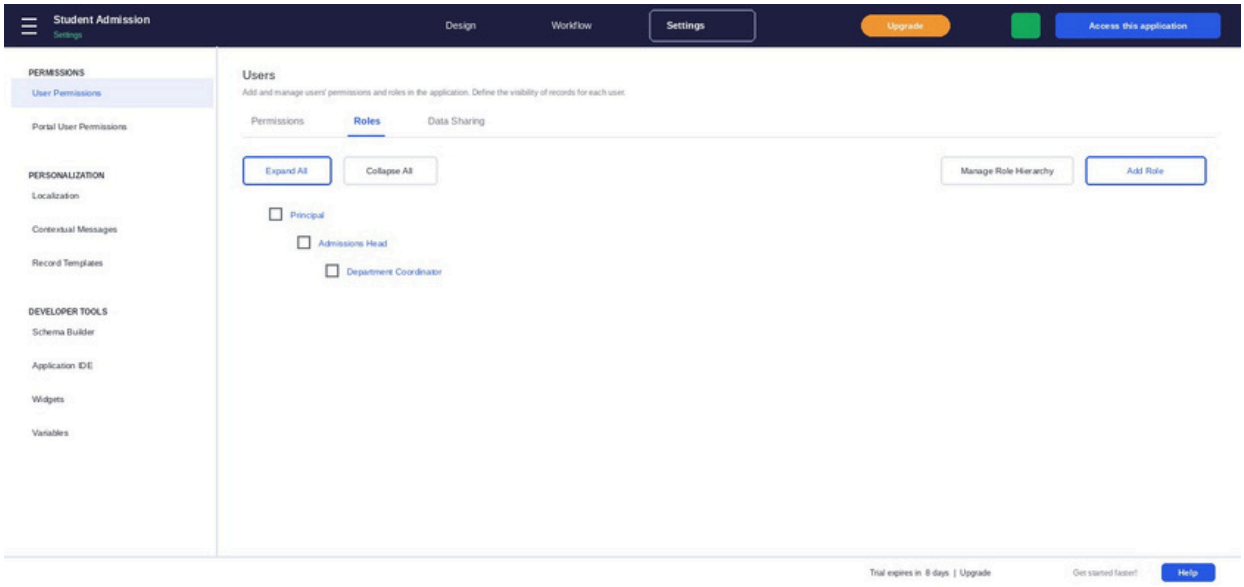


Figure 3: Roles and Role Hierarchy in the Student Admission application

The roles shown in the hierarchy are summarised below.

Role	Reports To	Position in Hierarchy
Principal	–	Top level of the hierarchy
Admissions Head	Principal	Second level
Department Coordinator	Admissions Head	Third level (lowest role shown)

Arranging roles in this way gives the application a clear reporting structure. The position of a role in the hierarchy influences which records its members can see, and further access can be given to a specific role through data sharing rules, as described in the next section.

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Data Sharing

Data sharing rules control the visibility of the records of a form. A rule specifies the form, the role or user from which records are shared, the role or user to which they are shared, and the level of access given. In this way, records can be made visible to additional roles or users beyond the default visibility of the role hierarchy. A data sharing rule was configured with the following details:

Parameter	Value
Rule Name	Student_Admission_Form
Form	Student Admission
Records Shared From	Roles – Admissions Head
Records Shared To	Roles – Department Coordinator
Level of Access	Read Only

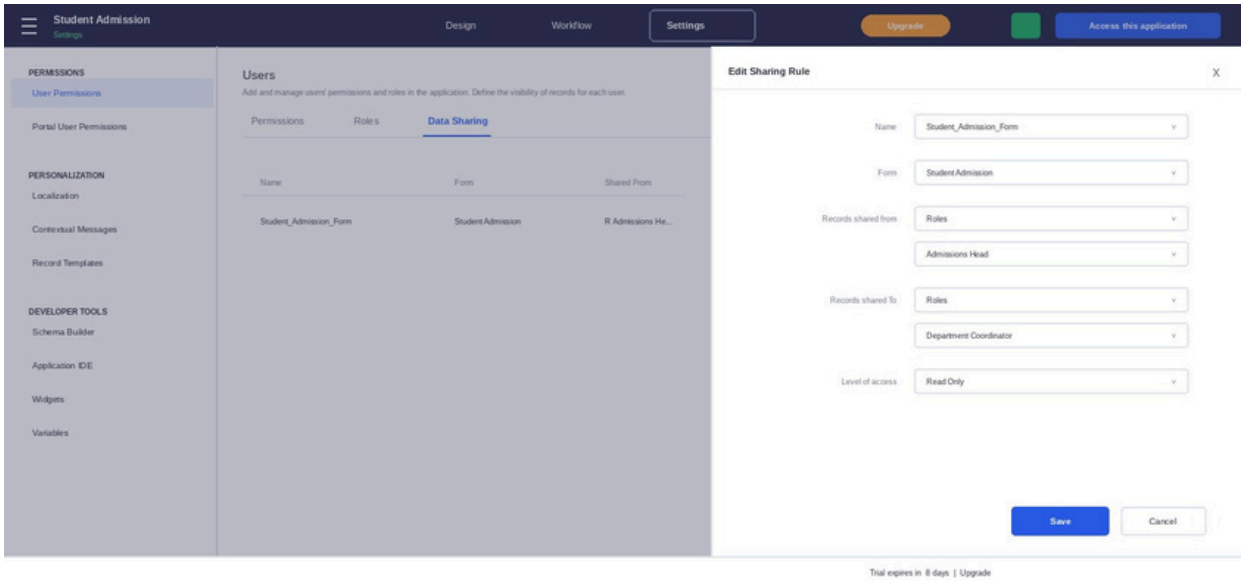
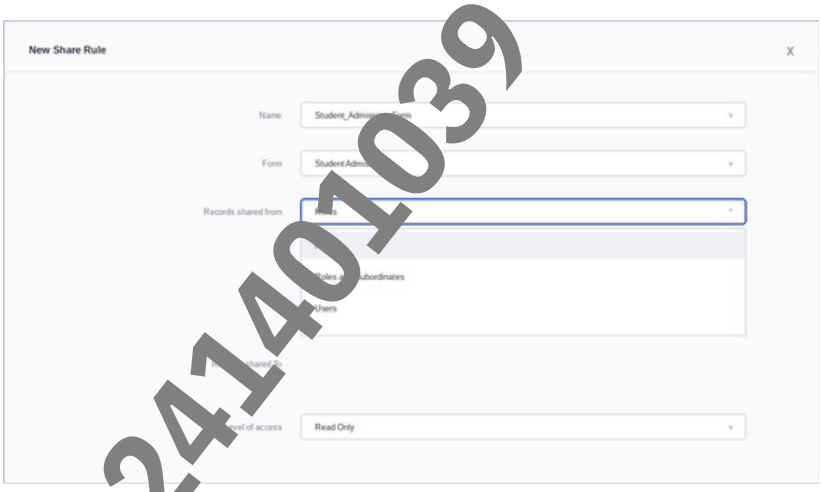


Figure 4: Edit Sharing Rule panel showing the Student_Admission_Form rule

Figure 4 shows the saved rule in edit mode; the rule also appears in the Data Sharing list in the background with its name, form and the shared-from role. While creating a rule, the New Share Rule panel offers three choices under Records shared from (Figure 5).



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Figure 5: New Share Rule panel – options for Records shared from

The three options are Roles, Roles and Subordinates, and Users. Roles shares the records of a single selected role, Roles and Subordinates shares the records of a role together with the roles beneath it in the hierarchy, and Users shares the records of selected individual users. In this experiment Roles was selected, with the Admissions Head role as the source. With Read Only access, members of the Department Coordinator role can view the shared records of the Student Admission form but cannot modify them.

Security Features and Governance Controls

Figure 6 is a conceptual diagram that classifies the security and governance controls of Zoho Creator as they apply to the Student Admission application. It is not a screenshot, and it is not proof that every listed feature was configured.



Figure 6: Classification of Security and Governance Controls (Conceptual Diagram)

Control	Explanation
User access control	User permissions (Read, Write and Admission_Officer_Access), role-based access and portal user permissions determine who can use the application and what they can do.
Data governance	Data sharing, access level, record visibility and role hierarchy govern how application data is accessed.
Application governance	Settings such as localization, contextual messages, record templates, schema builder, application IDE, widgets and variables are used to administer and customise the application.
Role hierarchy	Arranges roles in a reporting structure that influences which records a role can see.
Data sharing	Extends the visibility of records from one role or user to another with a chosen access level.
Record visibility	Decides which records are visible to which users, based on roles and sharing rules.
Administrative controls	Allow an administrator to manage permissions, roles and application settings.

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In this practical, user permissions and the role hierarchy were examined, and a data sharing rule was configured. The other items in the diagram, such as portal user permissions, localization, schema builder and widgets, are available in the Settings menu of the application, but they were not configured in this experiment.

Portal User Permissions

Portal User Permissions is listed beside User Permissions in the Settings menu and is used to control the access of external portal users. It was not configured in this experiment, and no portal permissions were set for the Student Admission application.

Role-Based Access and Data Sharing Flow

Figure 7 is a conceptual diagram that combines the role hierarchy of the application with the data sharing rule that was configured. On the left, the roles Principal, Admissions Head and Department Coordinator are shown in reporting order. On the right, the rule Student_Admission_Form is traced from the Student Admission form, through the role it is shared from, to the role it is shared to and the level of access given.

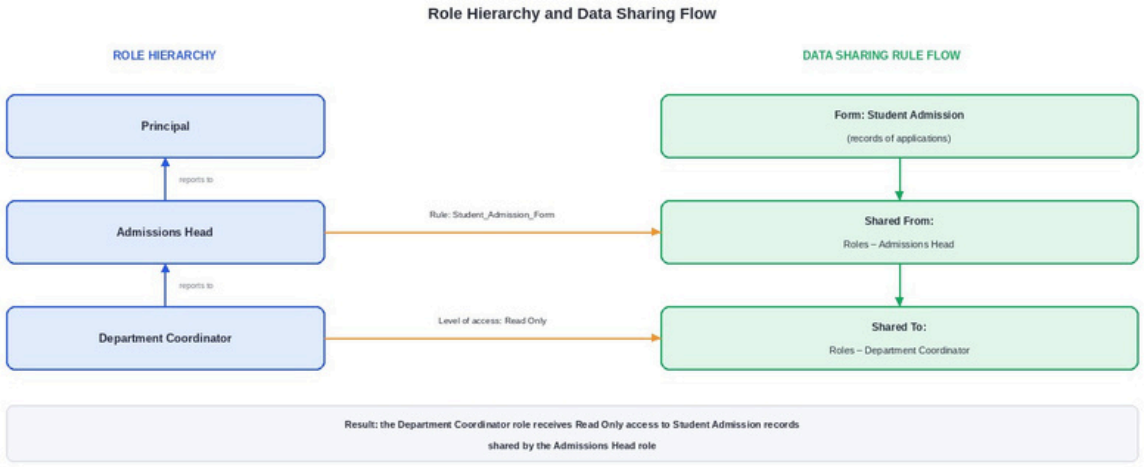


Figure 7: Role Hierarchy and Data Sharing Flow (Conceptual Diagram)

- The roles of the application are arranged in a hierarchy: Principal, Admissions Head and Department Coordinator.
- The data sharing rule Student_Admission_Form is defined for the Student Admission form.
- Records are shared from the Admissions Head role to the Department Coordinator role.
- The level of access is Read Only, so the Department Coordinator role can view the shared records but cannot modify them.

Governance Observations

- Predefined Read and Write permissions were available in the application, together with the Admission_Officer_Access profile.
- A data sharing rule named Student_Admission_Form was configured for the Student Admission form.
- The rule shares records from the Admissions Head role to the Department Coordinator role with Read Only access.
- Portal User Permissions was not configured in this experiment.
- The practical showed how access can be limited to what a role needs by choosing a suitable level of access.

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Conclusion

Zoho Creator provides built-in mechanisms for managing users, roles, permissions and data visibility. In this experiment, the Read, Write and Admission_Officer_Access permissions of the Student Admission application were examined, the role hierarchy of Principal, Admissions Head and Department Coordinator was viewed, and a data sharing rule named Student_Admission_Form was configured to share the records of the Student Admission form from the Admissions Head role to the Department Coordinator role with Read Only access.

The experiment shows how permissions, roles and data sharing work together so that each role sees the records it needs and no more. Choosing a suitable level of access, such as Read Only, allows records to be shared for collaboration without allowing them to be changed. Controlling who can access application data, and at what level, remains an essential part of building and managing business applications.